

Bi-Objective Vehicle Routing for Perishable Products Delivery With Consideration of Customers' Priorities and Customized Delivery Time Windows

Xinyue Liang, Haoxun Chen^{ID}, Nengmin Wang^{ID}, and Meng Zhang^{ID}

Abstract—As fresh product e-commerce experiences exponential expansion, the delivery of fresh products which are perishable is becoming a large part of road freight transportation. Transporting perishable products is a challenging task as it is difficult to preserve their nutritional value and freshness. To keep the freshness of such products in delivery, efficient transportation is critical. Besides, customers nowadays attach more importance to service flexibility, which contributes to customer satisfaction and is highly valued by suppliers. However, the service flexibility was seldom addressed previously in the delivery of fresh products. In this paper, we study a vehicle routing problem faced by a supplier who delivers perishable products to customers. We consider the service flexibility of the supplier who offers different delivery modes associated with differentiated time windows to customers. The supplier aims to maximize both its profit and customer satisfaction, where the customer satisfaction is measured by the weighted average quality loss of perishable products during delivery. This bi-objective vehicle routing problem with different customer priorities and delivery time windows is structured as a bi-objective mixed-integer linear programming model. To obtain the Pareto frontier of the model, an adaptive large neighborhood search (ALNS) algorithm based on the ϵ -constraint method and a multi-objective ALNS algorithm are developed with several improvement strategies proposed. The computational results conducted on modified Solomon's instances validate the model and prove the efficiency of the algorithms. Sensitivity analysis of the problem and the algorithms provide some managerial insights.

Index Terms—Customer satisfaction, customized delivery time windows, vehicle routing problem for perishable products, bi-objective optimization.

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I. INTRODUCTION

IN RECENT years, driven by the growing adoption of various online shopping platforms and a rapid shift in the consumer behavior from traditional to online shopping, we have witnessed a significant increase of the transactions of online business to consumer (B2C). Particularly, the online demand for perishable products is growing rapidly [1]. This phenomenon has led to a surge in the transportation activities for perishable products. However, due to their perishable nature, such products experience a transportation-induced loss ratio of 30% in numerous countries, with apparent adverse effects on both the economy and the environment [2].

In response to the growing demand for perishable products, e-commerce companies are striving to provide customers with more flexible services in order to improve customers' purchasing experiences, where a commonly provided service is attended home delivery (ADH). Moreover, the quality of perishable products is becoming a more concerned issue nowadays and it is closely tied to customer satisfaction. Given the deterioration characteristic of perishable products, service time window might be necessary for these products as it is better to hand them over to customers promptly [3]. In addition, offering a specific delivery time can prevent missed delivery [4]. For instance, most UK grocery retailers (e.g. Tesco, Morrisons, Sainsbury's) present their customers with the option to schedule the delivery of their orders within one-hour. FairPrice Online, an e-commerce platform of the Singapore supermarket chain, offers customers with next-day non-overlapping two-hour delivery.

A narrow time window for the ADH poses challenges for suppliers to ensure on-time delivery. Nevertheless, not all customers require an accurate and short delivery time. For example, certain groups of customers such as pensioners, students etc. might be willing to tolerate a relatively wide delivery time window in exchange for a reduced delivery fee [5]. Correspondingly, Tesco (one of the UK's largest retailers), introduced "Flexi Saver Windows", which offers customers four-hour time windows besides the regular one-hour time windows. The investigation of differentiated time windows has also been studied in the literature. The vehicle routing is an essential part of distribution activities. In recent years, a considerable volume of studies has emerged centered on vehicle routing problem (VRP) for perishable products.

Based on the fact that customers have different preferences for delivery service, this paper focuses on a perishable products supplier who offers two delivery modes based on next-day ADH to customers. Different delivery modes lead to different delivery fees, different delivery time windows and their associated weights, and minimum freshness level guarantee of the products delivered to customers. Besides, the possibility of flexible departure times for vehicles is considered. To solve the perishable products delivery problem, we formulate it as a bi-objective VRP with differentiated time windows and flexible departure times. The two objectives of the problem are maximizing the supplier's profit through minimizing its transportation costs and maximizing customer satisfaction through minimizing the weighted average quality losses of the perishable products delivered to customers.

The main contribution of this study is four-fold. Firstly, we develop a bi-objective VRP model with flexible vehicle departure time for the perishable products delivery, which was not considered in prior studies. Secondly, our proposed model considers different delivery modes, each associated with a particular type of time window, a service priority specified by a weight, and a minimum freshness level guarantee, which were not addressed in prior relevant studies. Thirdly, we develop two solution methods to solve the problem, one is ALNS algorithm based on the ε -constraint method (ALNS- ε) and the other is a new devised multi-objective ALNS algorithm (MO-ALNS). A comparison of these two algorithms is made. Furthermore, some improvement strategies for the two algorithms are proposed by utilizing a post-processing procedure and identifying a property of the problem. Finally, we perform sensitivity analysis and draw managerial insights.

The subsequent sections of this paper are organized as follows. A review of the related literature is conducted, and our work is positioned with respect to the existing literature in section II. In section III, we introduce the problem studied with underlying assumptions and its mathematical model. The procedures and specifics of the two algorithms are described in section IV. Data sets and parameter settings for the instances tested are given and the computational study of the algorithms is conducted to evaluate their performances in section V. In addition, the sensitivity analysis and a case study are performed. Finally, this paper ends with conclusions of our work and prospects for future research in section VI.

II. LITERATURE REVIEW

As the demand for perishable products continues to rise, the VRP for perishables has become increasingly important due to its ability to consider the perishable nature of products when delivering to customers. Below we review the relevant literature from three streams of research.

A. VRP for Perishables

Osvald and Stirn [6] developed a VRPTW model for vegetables distribution and time-dependent travel-times were taken into consideration. To characterize the quality loss during the delivery of the products, a three-segment linear function was introduced by them, which is adopted to describe the

quality loss in our paper. We will illustrate this function in the next section. Liang et al. [7] presented a VRPTW model that considers delivering multiple types of perishable items and utilizing a refrigerated fleet. They proposed a branch-and-price method and an adaptive large neighborhood search algorithm to solve the model. Liu et al. [8] studied an electric vehicle routing problem for perishables to minimize total cost including vehicle cost, energy consumption and delayed delivery cost. An adaptive hybrid ant colony optimization method was applied to solve the problem.

Economic perspective is the main focus of single-objective vehicle routing for perishables [9], [10], [11], [12], [13], where only one solution is provided for the decision maker generally. In contrast, multi-objective vehicle routing is capable of offering a set of solutions based on trade-offs among multiple objectives, for instance, to incorporate the environmental factors [7], [14], [15], [16], customer satisfaction [17], [18], [19], [20], and other factors [15], [21], [22], [23], [24].

B. VRP for Perishables Considering Customer Satisfaction

In recent years, researchers have started considering customer satisfaction in the VRP for perishables, as it has become a critical factor for business success. Customer satisfaction means meeting the needs and expectations of customers in a given service period.

One of the main factors contributing to customer satisfaction is the time window constraint for each order delivery [25]. A single-objective VRP model for transporting perishable goods via refrigerated vehicles was formulated by Song and Ko [26], aiming to maximize customer satisfaction by minimizing the total waiting time. Rahami et al. [27] proposed a VRP model for distributing fresh food over multiple periods in a bi-objective framework. Their study promoted customer satisfaction by reducing the overall late deliveries. Ghanadpour et al. developed a three-objective dynamic VRPTW model with the goal to minimize transportation costs, minimize number of utilized vehicles and maximize customer satisfaction in terms of the fitness of delivery time with customers' schedules [28].

Another important factor contributing to customer satisfaction is the condition of the perishable products. Amorim et al. [29] proposed a bi-objective model to minimize costs while simultaneously maximizing the mean remaining lifetime of the products. Zhao et al. [30] formulated a time-dependent VRP model for perishables delivery by considering the congested road with the objective to achieve a reduction in both total delivery costs and customers' dissatisfaction. A multi-objective evolutionary algorithm (MOEA) was employed to tackle this problem. Wang et al. [31] defined a customer satisfaction function that takes into account both of the aspects mentioned above. They formulated a bi-objective model considering stochastic demand and integrated the NSGA-II algorithm with the ALNS to address the problem.

C. VRP for Perishables Considering Customer Priorities

Service priority for each customer has been considered as an important pillar to achieve better customer satisfaction.

We classify these studies into two categories, where differentiated customer priorities are embodied in: (1) multiple prioritized delivery time windows; and (2) orders assigned with different importance.

For the first category, there exist two ways to respect the delivery time windows with different priorities both in real industries and literature. One way is to grant customers the option to choose more than one available time windows. Beheshti et al. [32] developed a bi-objective VRP model considering multiple prioritized time windows (VRPMPTW) aiming at minimizing total travel costs and maximizing customer satisfaction. Specifically, customers have preference ranks for different delivery windows and are segmented into three groups with three different levels of priority. The problem was solved by a quantum-genetic meta-heuristic. Baradaran et al. [33] studied a bi-objective VRPMPTW in the same context of customers having varying preferences for different time windows, whereas they didn't group customers but considered stochastic travel times. Another way is to provide customers with time windows of varying duration. Retailers in the US, such as Peapod, have already implemented a system of flexible delivery windows, providing customers with multiple time windows of different lengths. Typically, they adopt a fixed charging scheme for delivery fees determined by different time windows, which may include discount for looser delivery windows [5]. To achieve the highest number of customer orders accepted and provide as many as possible customers with the option of choosing short time window, Kohler et al. studied a VRPTW in the context of flexible time window management. Two sets of time windows with different lengths were offered and a hybrid heuristic was utilized [34].

For the second category, Ma et al. [35] devised a single-objective VRPTW model for perishable products with the goal to maximize supplier's profits and a hybrid meta-heuristic was implemented. In their study, the total profit is defined as the profit obtained from serving customers minus the total weighted loss of rejected orders. The weight assigned to each customer directly affects the profit. Wang et al. [19] established a bi-objective VRPTW model taking both transportation cost and customer satisfaction into account for perishable product delivery, and the quality loss is formulated by a time-dependent exponential function. The priority of each customer is directly related to its demand and distance from the depot. Customers with higher priority will be delivered before others in the same route.

With respect to the relevant literature reviewed above, the VRP we study has several new features. Firstly, unlike setting zero departure time for each vehicle in most VRPs for perishables studied, we allow flexible departure time (not necessarily zero) for vehicles. Despite increasing the complexity of the routing problem, it offers a notable reduction in the degradation in the quality of perishable products during delivery. Secondly, except transportation costs, we also consider customer satisfaction, where two types of customers correspond to two delivery modes with different priorities and customizable time windows for delivering their orders. To the best of our knowledge, bi-objective vehicle routing for perishables with the consideration of customers' priorities, differentiated

delivery time windows and flexible departure time has not been studied in the literature. To solve the proposed bi-objective VRP, we develop two meta-heuristic methods to approximately obtain its Pareto front quickly, one is MO-ALNS and the other is ALNS- ε . The results from our numerical experiments demonstrate the efficiency and effectiveness of both methods.

III. PROBLEM DESCRIPTION AND FORMULATION

A. Problem Description

We examine a vehicle routing problem where a supplier delivers perishable products to customers within a finite planning horizon. The supplier owns a certain number of homogeneous vehicles, knows the location, the demand and the delivery mode of each customer, and wants to simultaneously maximize its profit and customer satisfaction, which can be equivalently transformed into simultaneously minimizing its transportation cost and weighted average quality loss of the perishable products during the delivery.

This problem can be described in a non-directed road graph denoted by $G = (V, E)$, where $E = \{(i, j) | i, j \in V\}$ denotes the edges set and $V = \{0, 1, \dots, n\}$ denotes the set of nodes. In V , 0 represents the depot and $1, 2, \dots, n$ represents the customers. The finite planning time horizon of the supplier is T_H . The supplier provides customers with two delivery modes, denoted by set $M = \{1, 2\}$, where mode 1 imposes a hard delivery time window specified by a customer, and mode 2 has a looser delivery time window restricted only by the latest allowed delivery time specified by a customer. These two delivery modes lead to two delivery fees and all customers are grouped into two classes according to their delivery mode choices. These two types of customers are associated with two different weights indicating their service priorities. The customers who pay higher fees have a higher weight (priority) and are guaranteed to receive their ordered products with a higher minimum freshness level.

B. Assumptions and Notations

1) *Assumptions*: (1) The vehicle fleet initially located in the depot is sufficient to complete all delivery tasks.

(2) Each customer is served by one vehicle for once.

(3) No split deliveries are permitted, and the demand of each customer is within each vehicle's capacity.

(4) The perishable product is at its freshest level in the depot, and its quality will deteriorate during the delivery.

2) *Deterioration Function*: Fruits and vegetable products experiencing zero-order reaction of degradation has been proved by Labuza [36]. Customers will perceive the quality of a perishable product and place an intangible value on its relative freshness [37]. To explicitly describe the real and apparent quality loss of a fresh vegetable, a piecewise linear function defined on three time periods was proposed by Osvald and Stirn [6], where the first period is called the apparent stability period, during which the quality of the perishable product is decreasing, while this quality deterioration cannot be visually perceived. A perceptible quality deterioration occurs during the second period, and the product becomes inconsumable at the beginning of the third period. The perishable product

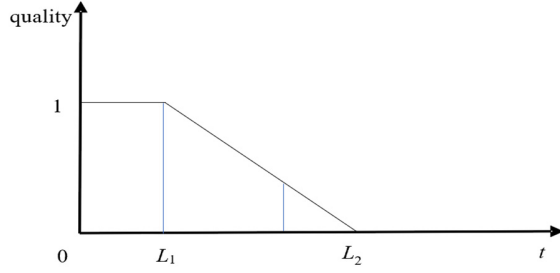


Fig. 1. Quality deterioration of a perishable product.

considered in our problem is fresh vegetables. We adopt the three-period piecewise deterioration as shown in Fig.1 and the meanings of L_1 and L_2 will be explained in the following subsection.

3) *Notations*: Based on the preceding problem description, the following symbols are used in the mathematical model.

Decision variables:

$x_{ij}^k = 1$, if vehicle k travels edge (i, j) ; otherwise $= 0$

$y_i^k = 1$, if vehicle k delivers the perishable product to customer i ; otherwise $= 0$

t_i^k the starting service time at customer i by vehicle k , t_0^k is the departure time of vehicle k from the depot

tr_0^k the return time of vehicle k to the depot

β_i the quality loss of the perishable product received by customer i

td_i the time duration of the quality loss process during transportation

Parameters:

N Set of all the customers nodes, $N = N_1 \cup N_2$

K Set of vehicles, $k \in K = \{1, 2, \dots, p\}$

M Set of delivery modes, $m \in M = \{1, 2\}$

W Set of customer weights, $w \in W = \{w_m | m = 1, 2\}$

R Set of guaranteed minimum freshness levels, $r \in R = \{r_m | m = 1, 2\}$

e_i the earliest service starting time of customer i

l_i the latest service starting time of customer i

s_i the service time at customer i

d_{ij} the length of edge (i, j)

q_i the demand of customer i

t_{ij} the edge (i, j) travel time

Q the maximum capacity of each vehicle

T_H the time duration of the whole planning horizon

c the cost of transportation per unit distance for each vehicle

c_f the fixed startup cost per vehicle

L_1 the time when the noticeable quality loss starts

L_2 the time when the shelf life of the perishable product ends

C. Mathematical Model

$$f_1 = \min \left(c_f \sum_{k=1}^p \sum_{j=1}^n x_{0j}^k + c \sum_{k=1}^p \sum_{i=0}^n \sum_{j=0}^n x_{ij}^k d_{ij} \right) \quad (1)$$

$$f_2 = \min \sum_{m=1}^2 \sum_{i \in n_m} w_m q_i \beta_i \bigg/ \sum_{i=1}^n q_i \quad (2)$$

$$\text{Subject to: } \sum_{h=0}^n x_{hi}^k - \sum_{j=0}^n x_{ij}^k = 0 \quad i \in V, k \in K \quad (3)$$

$$\sum_{k=1}^p y_i^k = 1 \quad i \in N \quad (4)$$

$$y_i^k = \sum_{h=0}^n x_{hi}^k \quad i \in N, k \in K \quad (5)$$

$$\sum_{i=1}^n y_i^k q_i \leq Q \quad k \in K \quad (6)$$

$$e_i \leq t_i^k \leq l_i \quad i \in N, k \in K \quad (7)$$

$$t_i^k + t_{ij} + s_i - M_0 (1 - x_{ij}^k) \leq t_j^k \quad i \in V, j \in N, k \in K \quad (8)$$

$$t_i^k + s_i + t_{i0} - M_0 (1 - x_{i0}^k) \leq tr_0^k \quad i \in N, k \in K \quad (9)$$

$$t_0^k \geq 0 \quad k \in K \quad (10)$$

$$tr_0^k \leq T_H \quad k \in K \quad (11)$$

$$td_i \geq 0 \quad i \in N \quad (12)$$

$$td_i \geq t_i^k - t_0^k - L_1 - M_0 (1 - y_i^k) \quad i \in N, k \in K \quad (13)$$

$$\beta_i = td_i / L_2 - L_1 \quad i \in N \quad (14)$$

$$1 - \beta_i \geq r_m \quad m \in M, i \in N_m \quad (15)$$

$$x_{ij}^k, y_i^k \in \{0, 1\} \quad i \in N, k \in K \quad (16)$$

Objective function (1) maximizes the supplier's profit through minimizing its total transportation cost. Objective function (2) minimizes the weighted average quality loss of the perishable products delivered to customers in order to maximize customer satisfaction. Constraint (3) is the flow conservation equation. Constraint (4) means that each customer can receive service only once from one vehicle. Constraint (5) ensures that each vehicle is restricted to serving customers assigned to its specific route. Constraint (6) imposes a restriction on the capacity of each vehicle. M_0 is a constant that surpasses any feasible route's overall travel time. Constraint (7) is the time window constraint. Constraint (8) and (9) indicate that any two nodes that are consecutively visited in the same route must have a compatible visit time. All vehicle routes must be conducted within the planning horizon, as stipulated by constraint (10) and constraint (11). Constraints (12)-(14) describe the quality deterioration process illustrated in Fig.1, where constraints (12)-(13) are used to calculate the deterioration time of the product delivered to each customer during transportation and constraints (14) calculate the corresponding quality loss of the product delivered to each customer. Constraints (15) ensure that the remaining quality of the perishables delivered to each customer meet the minimum freshness level requirement. Constraints (16) ensure all decision variables are binary integer.

IV. SOLUTION METHOD

VRP has been proved as an NP-hard problem [38]. Since VRP is embedded in our proposed problem, we develop

two meta-heuristics based on adaptive large neighborhood search (ALNS), one is ALNS based on the ε -constraint method (ALNS- ε) and another is the multi-objective ALNS (MO-ALNS).

ALNS, firstly introduced by Ropke and Pisinger in 2006 [39], is a powerful and versatile local search based heuristic which has been widely used to solve a large variety of optimization problems especially vehicle routing problems [7], [40], [41]. ALNS realizes large neighborhood search by applying various destroy and repair operators, empowered with an adaptive mechanism which updates the weight of each operator according to its historical performance during the search process. The probability of each operator to be selected to apply in the next iteration depends on its weight. A simulation-annealing based mechanism is used to determine whether a new solution found in each iteration is accepted. For multi-objective optimization problems (MOOP), various solution methods have been proposed in the literature such as the weighted sum method, the ε -constraint method and evolutionary algorithms. Evolutionary algorithms such as NSGA-II, SPEA-II and MO-PSO are highly favored in addressing the MOOP [42]. In particular, NSGA-II has achieved widespread adoption and numerous variants have been developed to enhance its performance, with alternative ranking schemes of Pareto fitness [43], various integrated strategies to improve convergence and population diversity [44] and different chromosome representations tailored to the specific nature of the problem [45] etc., all aimed at improving its applicability and effectiveness. The weighted sum method and the ε -constraint method are widely employed in MOOP. Between them, the ε -constraint method is more popular and has been widely used to solve MOVRPs. The ε -constraint method, which can obtain the entire pareto front [46], [47], was firstly introduced by Haimes et al. in 1971 [48]. By converting all but one objective of a MOOP into constraints, this method transforms the problem into a series of single-objective optimization problems [49]. In addition, an algorithm named MO-ALNS was recently proposed by Rifai et al. [50] to solve a multi-objective flow shop scheduling problem, it implements ALNS in a multi-objective framework by performing the neighborhood search that modifies non-dominated solutions. Cota et al. [51] applied MO-ALNS to solve a green machine scheduling problem. Aydemir-Karadag [52] firstly applied MO-ALNS to solve a periodic location inventory routing problem in healthcare waste treatment. The comparison of MO-ALNS with ALNS and MO-LNS on multiple performance metrics showed that MO-ALNS achieved better performance.

From the above analysis, we believe that the combination of the ε -constraint method and ALNS should work well for our studied problem and MO-ALNS is well adapted to it. This motivates us to develop two solution methods ALNS- ε and MO-ALNS for the problem. Before presenting the two methods, we present a procedure for constructing an initial solution of the problem, which will be improved by ALNS- ε or MO-ALNS to obtain the Pareto front of the problem. Finally, a post-processing method is applied to eliminate dominated solutions.

Algorithm 1 Construction of an Initial Solution

Input: $G = (V, E)$, K , sorted list of all customers $\{1, 2, \dots, n\}$

Step1. Take customer i in the sorted list randomly, $i \in \{1, 2, \dots, n\}$

Step2. Create the first and empty vehicle route and set $k = 1$ (k is the index of the latest created vehicle route, k_p denotes the current insertion position of route k)

Step3. For the customer list, execute the following steps for each customer j , $j = i, i + 1, \dots, n, 1, \dots, i - 1$:

- (a) Check the vehicle capacity feasibility of route k if insert j to position k_p : If not feasible, update $k = k + 1, k_p = 1$; else, turn to 3(b);
- (b) Check the time window feasibility of route k if insert j to position k_p : If not feasible, update $k = k + 1, k_p = 1$; else, turn to 3(c);
- (c) Perform the insert of customer j in route k with the smallest cost increment.

Step 4. Check the minimum perishability feasibility: If not feasible, go to step 5; otherwise, output the solution and end.

Step 5. Remove and reinsert those customers who cause the infeasible solution. The reinsertion procedure follows the rules from step1 to step 5.

Output: The initial solution x .

A. Initial Solution Construction

The location of each customer node is given by its Euclidean coordinate. All customers are first ranked in ascending order of their delivery time windows $(3l_i - e_i)/2$, and customers having equal values of this term are arranged in descending order of their distance from the depot. To obtain a feasible initial solution, we make use of the mechanism proposed by Wang et al. [53] to update the earliest and latest service time allowed by each customer to satisfy its delivery time constraint. The following Algorithm 1 constructs the initial solution.

Potential insertion positions of customer j should be limited in those positions where the inequality $t_i^k \leq l_j$ is satisfied, to help narrow the search space so as to build the initial solution quicker.

B. ALNS Based on ε -Constraint Method

1) ε -Constraint Method and ALNS: To implement the ε -constraint method, we transform the second objective into a constraint, and the derived single-objective optimization problem is formulated as follows:

$$\text{Problem } P' \min f_1(x) \quad (17)$$

$$\text{s.t. constraints (3) – (16)} f_2(x) \leq \varepsilon \quad (18)$$

where ε is the upper bound of the value of $f_2(x)$.

To obtain the Pareto front of the bi-objective problem studied, the value of ε is decreased by a small value δ at each iteration, thus it is essential to know the range of ε . The range is bounded by the ideal point (f_1^I, f_2^I) and the nadir point (f_1^N, f_2^N) . Let ε^L represents the lower bound value and

ε^U represents the upper bound value of ε . Firstly, we solve problem P_1 to obtain the value of f_1^I .

$$\text{Problem } P_1 f_1 = \min \left(c_f \sum_{k=1}^p \sum_{j=1}^n x_{0j}^k + c \sum_{k=1}^p \sum_{i=0}^n \sum_{j=0}^n x_{ij}^k d_{ij} \right) \quad (19)$$

s.t. constraints (3) – (16)

Then we can obtain the value of f_2^N which equals the value of ε^U and it is obtained by solving problem P_2 .

$$\text{Problem } P_2 f_2 = \min \sum_{m=1}^2 \sum_{i \in n_m} w_m q_i \beta_i \Big/ \sum_{i=1}^n q_i \quad (20)$$

s.t. constraints (3) – (16), $f_1 = f_1^I$

Similarly, the value of ε^L can be obtained by swapping the position of f_1 and f_2 . Moreover, a property of the problem, given by the following proposition, is derived, which can help reduce the computational complexity for calculating ε^L to $O(n)$.

Proposition 1: The second objective's lower bound of the problem is derived by its solution in which all customers' orders are delivered and each vehicle only serves a single customer order.

Proof: Let S denotes the solution. We demonstrate this property through a proof by contradiction. Assume a smaller lower bound exists for the second objective with the corresponding solution S' . Then, at least one route within this solution must involve a vehicle delivering more than one customer order. Suppose that one such route is $p' = \{0, x_1, x_2, 0\}$ which serves two customers x_1 and x_2 . The time of the vehicle's arrival at customer x_2 in this route is not less than that at the same customer x_2 if it is the only customer served individually in the route $p = \{0, x_2, 0\}$. Hence, the deterioration in quality of the perishable products upon receipt by customer x_2 in solution S' will be the same or more than in solution S defined in this proposition, resulting in a value for the second objective that is equal to or greater, contradicting the assumption. This ends the proof.

Each individual problem P' is solved using ALNS outlined in Algorithm 2, where ΔE is the variation in the objective function (2) values between two solutions.

2) *Destroy and Repair Operators:* This subsection describes the operators used in the ALNS algorithm. To improve existing solutions and balance between diversification and intensification in neighborhood search, five destroy operators and four repair operators are designed and used in the ALNS algorithm.

a) *Destroy operators:* **O1 Random removal:** This operator randomly removes h customer nodes, which help to avoid trapping in local optimum.

O2 Shaw removal: Such a removal was firstly proposed by Shaw. The general idea of this removal is to remove several similar customer nodes to some extent, because there is a higher probability that similar customer nodes will be reinserted in new locations, perhaps resulting in a better solution. To implement this removal, we have to define a relatedness measure $R(i, j)$ for any pair of customer node i and j . For

Algorithm 2 ALNS Algorithm

Input: an incumbent non-dominated solution x , the set of operators $\Omega = \Omega^- \cup \Omega^+$, the set of scores of operators $\{\pi_i | i = 1, 2, \dots, |\Omega|\}$, the size of a segment λ , and initial value T_0 of temperature T

Repeat:

Step 1: Select a destroy operator $d \in \Omega^-$ and a repair operator $r \in \Omega^+$ based on the selecting scheme;
 Step 2: Apply the destroy and repair operator successively to solution $x: x' = r(d(x))$; compute $\Delta E = f(x') - f(x)$
 if $\Delta E < 0$ then
 $x = x'$, update π_d and π_r by σ_1 ;
 elif $\Delta E = 0$ then
 $x = x'$, update π_d and π_r by σ_2 ;
 else generate a random number r from uniform distribution $U[0,1]$, if $r < \exp(-\Delta/T)$ then
 $x = x'$, update π_d and π_r by σ_3 ;
 End if
 Step 3: $T = T\alpha$, $iter = iter + 1$
 Step 4: if $iter = \lambda$, update the weights of operators and reset their scores (See equations (25) and (27) below)

Until the stop criterion is met

End while

Output: A new non-dominated solution x

Algorithm 3 Shaw Removal

Input: choose a customer node a from a solution x at random, the set of removed nodes $D = \{a\}$;

While $|D| < g$ **do:**

 Randomly select a customer node i in D ; B is the set containing all nodes in x but not in D

for each customer node j in B :

 Calculate $R(i, j)$ defined by equation (21) and sort them in non-decreasing order;

 Randomly generate a number b from the interval $[0,1]$;
 $D = D \cup \{[|B| \cdot b^l]\}$, $B = B / \{[|B| \cdot b^l]\}$;

End while

Output: Set D consists of g customer nodes removed from solution x

our problem, this measure is defined below by considering distance, delivery time window, and demand.

$$R(i, j) = \chi \cdot d_{ij} + \gamma \cdot (|e_i - e_j| + |l_i - l_j|) + \psi(|q_i - q_j|) \quad (21)$$

where χ , γ and ψ are three weight coefficients whose setting will be discussed in the next section. One parameter for this operator is its degree of randomization denoted by l . Smaller values of l are associated with higher randomness. Algorithm 3 outlines the steps for implementing the Shaw removal procedure.

O3 Least Route Removal: This operator is designed to remove customers from the smallest route, contributing to reducing the total number of vehicles utilized. The size of each route is measured by two factors including its load ratio α_k ,

calculated by its current load divided by its maximum capacity, and its travel time ($tr_0^k - t_0^k$), as defined in equation (22). The route k with the smallest value of U_k is removed.

$$U_k = \alpha_k + (tr_0^k - t_0^k) \quad (22)$$

O4 Worst Cost Removal: This operator aims to remove the first h customer nodes with the highest insertion cost $_1(i, s)$. (i, s) represents that customer i receives service from vehicle in solution s and equation (23) provides the definition of cost $_1(i, s)$, where $f_1^{-i}(s)$ is the value of f_1 when customer node i is removed from solution s .

$$\text{Cost}_1(i, s) = f_1(s) - f_1^{-i}(s) \quad (23)$$

O5 Worst Freshness Removal: The operator is designed to remove the first h customer nodes with the highest level of quality deterioration. The quality deterioration cost $_2(i, s)$ is defined as follows.

$$\text{Cost}_2(i, s) = w_m q_i \cdot \beta_i \quad (24)$$

b) Repair operators: In this subsection we describe four repair operators which aims to generate improved solutions. For the customer nodes removed from the current solution, the repair operators insert them sequentially into a new position in an existing partial route. The new position may be situated either within the same route but at a different location, or a position in a different route. The insertion process involves the feasibility check and recalculating the values of the two objectives. The details of these operators are given below.

O6 Cheapest cost insertion: This operator searches for the best insertion position with least cost increase for each customer node to be inserted. Let $\Delta f_{s'}(i, k)$ denote the cost change induced by inserting customer i into vehicle route k in solution s' . We define the cost change as the variation in the value of objective function (1). In each iteration, after obtaining the best route and the best position for inserting a customer i^* that yields the least cost increase, the best insertion route is the only route that needs to be updated. A new route is initiated if it is impossible to feasibly insert the customer into any existing route. The repair operator is described below.

O7 Greedy Cheapest Insertion: For each removed customer node j , this operator is designed to insert it into one of the k selected routes at the best position. The route selection criterion is determined by the waiting time, vehicle departure time and load ratio. The route satisfying the inequality $t_0^k < l_j$ with the longest waiting time is the final chosen route. The insertion step is similar to that in algorithm 4. In case the feasibility is violated by this insertion, a new route will be created in order to insert the customer.

O8 Cheapest Distance Insertion: This operator seeks the best insertion position among all the routes in the incumbent solution s' for each customer node to be inserted by considering the travel distance change. The procedure of this operator is almost the same as that of algorithm 4, except that $\Delta f_{s'}(i, k)$ is replaced by the travel distance.

O9 Cheapest Freshness Loss Insertion: This operator is designed to consider the second objective function. It explores feasible insertion positions in each route considering only the freshness loss. The process and steps utilized by this

Algorithm 4 Cheapest Cost Insertion

Input: solution s' , removed nodes set D

While $|D| > 0$ **do**

$\Delta f_{s'}(i^*) = \infty$; $\Delta f_{s'}(i, k) = \infty \forall i \in D$

for each customer $i \in D$:

$\Delta f_{s'}(i) = \infty$;

for $k \in s'$:

for potential position j **in route** k :

check the feasibility of inserting i to pos j in route k ,
if true:

perform the insert and calculate $\Delta f_{s'}(i, k)$;

if current $\Delta f_{s'}(i, k) < \text{original } \Delta f_{s'}(i, k)$:

perform the insertion of customer i , update $\Delta f_{s'}(i, k)$

remove i

if $\Delta f_{s'}(i, k) < \Delta f_{s'}(i^*)$:

Update $\Delta f_{s'}(i^*) = \Delta f_{s'}(i, k)$, $i^* = i$ and save the best insertion position;

if $\Delta f_{s'}(i^*) = \infty$:

open a new route for i and insert it to the new route, update $D = D \setminus \{i\}$

Else

insert i^* to the best position, update $D = D \setminus \{i\}$

operator's algorithm adhere those of O8, the only difference is that the assessment of each insertion position is based on the freshness loss increment $w_m q_j \Delta t_{\text{increase}}$. Given k partial routes in the current solution with the number of customer nodes of route j represented as l_j , $j = 1, 2, \dots, k$, for each potential insertion position in route j , the feasibility check and freshness loss change computation both have a complexity of $O(l_j)$. Consequently, the complexity of inserting one customer in the best position for this operator is $\sum_{j=1}^k O(l_j^2)$

3) Adaptive Weight Adjustment and Operator Selection Scheme: In this subsection we present the details of how the weights of the operators are adjusted automatically and how an operator is selected based on the weights. The adaptive adjustment relies on the historical performance of each operator, quantified by the frequency of its utilization. The weights are updated after each segment iteration. A segment consists of λ consecutive iterations performed by the ALNS algorithm.

The score of every operator starts at zero as each segment begins, and it is updated after completing each segment. When a new solution achieves the global best, the scores of the selected destroy operator and insertion operator are increased by σ_1 . If the new solution is superior to the incumbent solution, the scores of the selected operators are increased by σ_2 . If the new solution is inferior but accepted, the scores of the selected operators are increased by σ_3 . Each operator's weight is equally set initially. Let $wo_{i,j}$ denote the weight of operator i after executing segment j , this weight is updated according to equation (25):

$$wo_{i,j+1} = (1 - u)wo_{i,j} + u\pi_i / \theta_{i,j} \quad (25)$$

where π_i is the score of operator i , $\theta_{i,j}$ is the number of iterations in which operator i is selected in j th segment and $u \in [0, 1]$ serves as a control parameter that determines how

large the segment span the weights are updated. The operator selection complies with the roulette wheel principle, and the probability for operator i to be selected in each segment is given by equation (26):

$$wo_{i,j} / \sum_{i \in \Omega^- \cup \Omega^+} wo_{i,j} \quad (26)$$

4) Solution Acceptance and Stop Criterion:

Kirkpatrick et al. first introduced the Simulated Annealing (SA) method and it has demonstrated successful applications to NP-hard problems [54]. The SA method allows acceptance of some inferior solutions with a probability in neighborhood search. As conventional ALNS, we also use SA as the solution acceptance method.

Suppose that a new solution denoted as x' is obtained by utilizing destroy and repair operations to incumbent solution x , and the corresponding objective values of the two solutions are $f(x')$ and $f(x)$, respectively. If x' is not better than x , within the interval $[0,1]$, a value r is randomly generated. Should r be smaller than the probability specified by $\exp(-(f(x') - f(x))/T)$, the current solution is updated with the new solution x' as shown in Algorithm 2. A linear cooling scheme $T = \alpha T$ is used in our ALNS algorithm to decrease the value of T in each iteration. The stop criterion of the algorithm is set as $T < 0.01$ or the number of iterations exceeds 1000.

C. MO-ALNS

Instead of converting the bi-objective optimization problem into a sequence of single-objective optimization subproblems with the evaluation of each solution made on the non-transformed objective, the MO-ALNS algorithm utilizes the multinomial probability mass function for objective selection in each iteration and each solution is then evaluated by the selected objective. This method has been used in several related studies and has been proved its effectiveness and agility in multi-objective optimization [55]. A conclusion was made by Satan's study that equal probability works the best [56]. Therefore, we set an equal selection probability (50%) to each objective in our proposed problem.

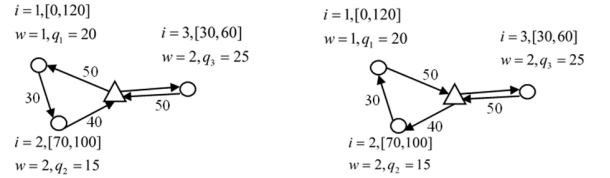
1) *Modified Adaptive Weight Adjustment and Solution Acceptance*: To incorporate ALNS into the bi-objective optimization framework, the weight adjustment mechanism is modified. The score and the weight of each operator now both depend on the objective selected. π_{io} is also initialized as zero at the beginning of each segment. Accordingly, the weight of operator i for objective o after executing segment j , represented by $wo_{i,o,j}$, is updated according to equation (27):

$$wo_{i,o,j+1} = \begin{cases} wo_{i,o,j} & \text{if } \theta_{ij} = 0 \\ (1-u)wo_{i,o,j} + u\pi_{io}/\theta_{ij} & \text{if } \theta_{ij} \neq 0 \end{cases} \quad (27)$$

2) *Non-Dominated Solutions Construction*: Another difference between MO-ALNS and ALNS lies in how the Pareto front is constructed. Here we adopt the AddSolution procedure proposed by Lust and Teghem [57]. This procedure has been successfully integrated in a MO-ALNS algorithm [49].

Algorithm 5 AddSolution procedure

Input: Set X consists of the set of non-dominated solutions X_n and a dominated solution s'' , current solution s'
 Randomly select a solution s from X
 if $f(s') < f(s)$
 delete s from X , add s' to X ;
 if $!(f(s') < f(s)) \wedge !(f(s') > f(s))$
 add s' to X ;
 elif $f(s') > f(s)$
 use SA scheme to determine whether to add s' to X
 Endif
Output: X



(a) solution before post-processing (b) solution after post-processing

Fig. 2. An example of post-processing.

The AddSolution procedure is applied to update the set of non-dominated solutions and simple Pareto dominance is applied to assess solutions. To further preserve randomness and prevent local optimum, we keep one dominated solution. The pseudocode of this procedure is outlined in Algorithm 5.

D. Post-Processing

Some dominated solutions may be generated in the process of problem-solving. Hence, we introduce a post-processing procedure to remove them timely to help achieve a reduction in the computation time. This post-processing operation has a computational complexity of $O(n)$. The pseudocode of the post-processing is given by Algorithm 6. Fig.2 gives an example of applying the post-processing procedure.

In this example, a solution x before post-processing consists of two routes, which are $x_1 = \{0, 1, 2, 0\}$ and $x_2 = \{0, 3, 0\}$, respectively, as depicted in Fig.2. (a). For this solution, we can calculate its total quality loss as $2700/L_2 - L_1$. Solution x' , which has two routes $x'_1 = \{0, 2, 1, 0\}$ and $x'_2 = \{0, 3, 0\}$ as shown in Fig.2. (b), is obtained after applying the post-processing procedure to x . The total quality loss of x' is reduced to $1500/L_2 - L_1$ while the transportation costs of these two solutions are the same. Besides, the post-processing procedure includes the updating of the departure time for each vehicle route, considering their flexible nature.

V. COMPUTATION STUDY

In this section, we conduct a computational study to verify our model and to evaluate the effectiveness of the proposed algorithms. For small instances, the bi-objective optimization model is solved using the ϵ -constraint method with all derived single-objective optimization models solved by GUROBI 10.0.1 with default settings. Firstly, the solutions

Algorithm 6 Post-Processing**Input:** A feasible current solution x For route k in solution x , do:reverse it record as route k' update the timetable of route k' and check the feasibilityif the feasibility of k' is true:replace k with k' ;

else:

keep route k ; $k = k + 1$;Until each k is checked in solution x **Output:** An improved solution x'

obtained by MO-ALNS are compared to those obtained by GUROBI on the small instances. Then, we compare MO-ALNS, ALNS- ε , and NSGA-II on large-size instances. All computational experiments were executed on a PC equipped with a 2.30-GHz processor and 16 GB RAM, operating under Microsoft Windows 11. All computation times were measured in seconds.

A. Data Sets and Parameter Setting

The instances we tested were generated based on the C-type and RC-type benchmark instances of VRPTW established by Solomon [58]. Since the tightness of time windows varies between two types of customers in our problem, the Solomon instances were solely utilized for generating customer locations and the time windows were randomly generated to adapt to our problem. The customers who select the first delivery mode are set with a hard time window with 30min duration, and all the other customers are set with a looser hard time window starting from time zero to their allowed latest delivery time.

The parameters of our algorithms are set in the following way. According to [39], a good initial value of temperature T should be set by considering the specific problem to be solved. Following this, the initial value T_0 of T is determined by examining the initial solution s of our problem obtained by algorithm 1, with the probability of accepting a solution with its performance 5% worse than that of the current solution set to 0.5. Thus, we set $T_0 = 0.05 f_1(s) / \ln 2$. Among the other parameters, the maximum removal numbers g and h for removal operators can significantly affect the solution quality [59], and the cooling rate α for temperature decrease affects the convergence of the algorithm. To tailor the values of the three key parameters to our problem, we conducted preliminary experiments on these key parameters and used Taguchi's method, which uses orthogonal arrays to reduce the number of experiments, to find their best values [60]. In our experiments, parameter g and h were both randomly generated with $4 \leq g \leq \min(100, \xi_1 n)$ and $4 \leq h \leq \min(100, \xi_2 n)$, where two parameters ξ_1 and ξ_2 were tuned using Taguchi's method. We consider three values (levels) for each parameter of ξ_1 , ξ_2 and α with all possible combinations of the three parameter values given by TABLE I. Each instance with $N=50$ was run five times. TABLE II provides the results, and the second row

TABLE I
COMBINATIONS OF VALUES OF THREE PARAMETERS

No.	ξ_1	ξ_2	α
1	0.30	0.30	0.99925
2	0.30	0.35	0.99975
3	0.30	0.40	0.99950
4	0.35	0.30	0.99975
5	0.35	0.35	0.99950
6	0.35	0.40	0.99925
7	0.40	0.30	0.99950
8	0.40	0.35	0.99925
9	0.40	0.40	0.99975

TABLE II
SUMMARY OF THE EXPERIMENTAL RESULTS

No.	1	2	3	4	5	6	7	8	9
GAP	0.87	0.94	0.96	0.93	0.90	0.92	0.94	0.93	0.95

shows the ratio of the worst and the best solutions obtained in the experiments, as measured by hypervolume. We can derive that $(\xi_1, \xi_2, \alpha) = (0.3, 0.4, 0.99950)$ yields the best results. For the remaining parameters, we follow Aydemir and Karadag's and Ropke's settings and set $(\chi, \gamma, \psi, \sigma_1, \sigma_2, \sigma_3, l, u) = (9, 3, 2, 33, 9, 13, 6, 0.7)$. Finally, we set $L_1 = 420\text{min}$, $L_2 = 20\text{min}$, $r_1 = 0.7$, $r_2 = 0.5$.

B. Numerical Results

1) *Comparison With GUROBI*: TABLE III provides seven performance indexes and computation time of the proposed MO-ALNS for the eight instances with $N=10$ to evaluate its performance compared with GUROBI. These eight instances were generated by randomly selecting customers from large-size instances.

The computation time of GUROBI and the MO-ALNS in TABLE III, denoted by t_1 and t_2 respectively, are their average time over fifteen executions. The last three columns of TABLE III show the Pareto-optimal solution coverage number of the MO-ALNS (PO-SCN) and the average deviations to the optimum solutions obtained using the exact method, in terms of two objectives (1) and (2). The average deviation of each objective is evaluated by the difference of this objective between the obtained solution to the nearest optimal solution in each Pareto set, denoted as DEV 1 and DEV 2 in TABLE III. They are 0.05% and 0.02% on average for each objective, demonstrating the effectiveness of the MO-ALNS.

The other four indicators, namely cardinality $|R_F|$ and $|A_F|$, hypervolume ratio H and average e-dominance Z , are used to verify the Pareto front obtained by MO-ALNS. The cardinality $|R_F|$ and $|A_F|$ are the total number of pareto solutions respectively obtained by GUROBI and the MO-ALNS. Greater cardinality indicates superior performance. The hypervolume is one of the most fundamental and popular-utilized indicators to assess the quality of a multi-optimization algorithm [61]. It refers to the area of the region dominated by the obtained solution set, computed as the sum of areas occupied by rectangles created using solutions from the Pareto front and the nadir point. A higher hypervolume value indicates a better

TABLE III
SEVEN PERFORMANCE INDEXES AND COMPUTATION TIME OF MO-ALNS FOR THE EIGHT INSTANCES WITH N = 10

Inst ance	N	$ R_F $	$ A_F $	H	Z	t_1	t_2	PO- SCN	DEV 1	DEV 2
1	10	11	11	1	1	449.79	4.23	11	0.00%	0.00%
2	10	9	9	1	1	516.43	5.17	9	0.00%	0.00%
3	10	10	10	1	1	447.62	4.45	10	0.00%	0.00%
4	10	12	12	0.995	1.002	517.97	4.89	11	0.15%	0.07%
5	10	12	12	1	1	458.61	4.39	12	0.00%	0.00%
6	10	13	12	0.978	1.011	598.06	5.78	12	0.26%	0.12%
7	10	10	10	1	1	448.85	4.56	10	0.00%	0.00%
8	10	8	8	1	1	420.65	4.02	8	0.00%	0.00%
avg		10.63	10.5	0.997	1.002	482.25	4.69	10.38	0.05%	0.02%

quality of a non-dominated set. The average e-dominance Z as shown in equation (28) and (29) quantifies the mean separation between the two Pareto sets obtained. As Z approaches 1, the two Pareto sets become more similar. The Pareto front obtained by GUROBI is the reference set, which means that the hypervolume ratio and e-dominance of $|R_F|$ is equal to 1 and 0 so they will not be given in TABLE III.

$$e(x') = \min_{(\varphi(x), w(x)) \in A_F} \max\left\{\frac{\varphi(x)}{\varphi(x')}, \frac{w(x)}{w(x')}\right\} \quad (28)$$

$$Z = \frac{1}{|R_F|} \sum_{(\varphi(x'), w(x')) \in R_F} e(x') \quad (29)$$

The results presented in TABLE III consistently point to the conclusion that the MO-ALNS performs nearly optimally whereas its computation time is much shorter than that of the exact method, which provides evidence of the meta-heuristic method's effectiveness and efficiency in dealing with small-size instances.

2) *Comparison of MO-ALNS and ALNS- ε* : We then apply the two proposed meta-heuristics, MO-ALNS and ALNS- ε , to eight instances with N=50 and eight instances with N=100. In the following three subsections, we first conduct an analysis of the operators used, then conduct a comparison of the results obtained by these two methods and NSGA-II respectively in terms of four indexes, finally analyze the effectiveness of the post-processing procedure.

a) *Analysis of the operators*: We evaluate the performance of each operator by recording its using frequency percentage. TABLE IV gives the results of sixteen instances. The results indicate that the operator performance varies between different instances. Averagely, among the destroys operators, O4, O1 and O5 achieved better performance, and O3 is the least essential. We analyzed that customers served on the same route tend to cluster geographically and have consecutive time windows, posing a challenge when attempting to reassign them to alternative routes. For the repair operators, O6 is used more frequently than others on average, while O7 is the least crucial operator. Overall, we conclude that O4, O1, O5 and O6 play important roles in achieving better solution quality of the metaheuristic.

b) *Performance of MO-ALNS and ALNS- ε* : A comparison is made among the proposed two methods and NSGA-II. NSGA-II is configured with a population size of 100, a crossover probability of 0.7, a mutation probability

TABLE IV
OPERATOR USAGE FREQUENCY

No	O1	O2	O3	O4	O5	O6	O7	O8	O9
9	22.2	17.7	15.5	22.9	21.7	30.7	20.0	26.1	23.2
10	19.5	18.0	15.8	23.6	23.2	28.9	20.7	26.3	24.0
11	22.7	18.8	15.1	24.8	18.6	32.1	18.8	23.8	25.4
12	21.1	17.0	15.1	24.6	22.3	33.2	18.7	24.3	23.8
13	21.3	17.3	16.0	24.2	21.3	31.7	19.5	25.8	23.0
14	23.4	17.1	14.7	23.7	21.1	30.9	18.3	27.3	23.5
15	23.6	18.2	14.6	24.0	19.7	33.6	17.7	23.5	25.3
16	22.1	17.2	14.7	25.6	20.4	31.1	18.5	24.8	25.7
17	20.9	18.2	16.5	23.0	21.4	30.8	18.1	24.0	27.1
18	18.9	18.5	16.9	23.8	22.0	29.8	19.4	25.7	25.0
19	19.1	16.7	15.6	24.3	24.4	32.2	19.8	24.8	23.2
20	21.1	17.0	16.3	25.2	20.5	32.4	18.2	23.7	25.7
21	22.7	19.7	14.2	24.2	19.2	33.1	17.6	23.2	26.1
22	23.2	19.1	14.4	25.1	18.3	34.1	16.8	22.8	26.4
23	21.0	15.9	16.2	23.7	23.2	32.3	18.5	29.4	19.8
24	19.7	15.6	18.1	24.1	22.5	31.4	18.9	24.9	24.9
avg	21.4	17.6	15.6	24.2	21.2	31.8	18.7	25.0	24.5

of 0.2, and a maximum iteration number of 1000 [62]. TABLE V shows the hypervolume and computation time of these three methods. It suggests that both ALNS- ε and MO-ALNS outperform NSGA-II in terms of the hypervolume on average. Among the sixteen instances, MO-ALNS outperforms ALNS- ε for seven instances and achieves a higher average hypervolume. The results indicate that both methods work well for the studied problem and the MO-ALNS can generate high-quality pareto solutions. According to Table V, when N, the number of customers, increases from 50 to 100, the average computation time of ALNS- ε , MO-ALNS, and NSGA-II increases from 300.49s to 1118.69s, 261.79s to 960.94s, and 203.67s to 894.96s, respectively. Combining the computation time of N=10, it can be derived that the computation time of each algorithm approximately follows a quadratic increase as the size of the problem increases. The computation time function can be fitted to $f_1(n) = 0.10134n^2 + 1.1628n$ for ALNS- ε and $f_2(n) = 0.08395n^2 + 1.3905n$ for MO-ALNS using polynomial interpolation method. Moreover, MO-ALNS has a shorter computation time compared to the ALNS- ε , with 14.10% saving on computation time.

We also use dominance percentage to evaluate the solution methods as shown in equation (30). Suppose the Pareto sets obtained by MO-ALNS, ALNS- ε and NSGA-II respectively are S_1 , S_2 and S_3 , we merge these three sets together forming a set $S = S_1 \cup S_2 \cup S_3$. Then the dominated solutions in each set

TABLE V
HYPERVOLUME AND COMPUTATION TIME FOR THE INSTANCES WITH N = 50 AND 100

No.	Hypervolume			Time(sec)		
	ALNS- ϵ	MO-ALNS	NSGA-II	ALNS- ϵ	MO-ALNS	NSGA-II
9	1890.24	1943.16	1853.92	278.05	255.70	193.26
10	1822.11	1794.78	1731.83	300.70	271.32	200.70
11	1951.88	1988.96	1881.89	316.80	265.30	208.78
12	1873.70	1851.21	1820.12	314.14	264.51	212.50
13	1805.30	1852.24	1749.08	303.12	256.90	211.66
14	1838.72	1855.26	1815.11	303.50	269.47	211.10
15	1921.84	1912.23	1862.95	293.23	263.31	206.08
16	1910.07	1864.23	1884.77	294.40	247.80	185.24
avg1	1876.73	1882.76	1824.96	300.49	261.79	203.67
17	6697.57	6798.04	6503.15	986.48	954.12	790.34
18	6533.20	6422.14	6439.65	1194.71	988.66	922.66
19	6611.66	6558.76	6401.96	1165.48	952.08	920.10
20	6595.88	6510.13	6264.62	1224.74	946.09	913.26
21	6618.54	6426.61	6375.71	1095.04	971.03	923.21
22	6490.04	6600.37	6267.23	1147.65	1013.89	922.25
23	6504.21	6972.51	6286.16	1086.41	939.22	863.96
24	6424.03	6327.67	6284.30	1049.01	922.46	903.92
avg2	6559.39	6577.03	6352.85	1118.69	960.94	894.96

TABLE VI
SOLUTIONS OBTAINED AND PERCENTAGE OF DOMINANCE

No.	Solutions Obtained			Dominance Percentage(%)		
	ALNS- ϵ	MO-ALNS	NSGA-II	ALNS- ϵ	MO-ALNS	NSGA-II
9	18	18	15	65.98	78.33	57.67
10	21	20	17	76.67	66.59	67.98
11	20	20	16	59.38	65.72	50.05
12	20	19	17	67.92	61.65	49.90
13	19	19	15	71.83	82.27	57.96
14	18	18	15	70.57	78.09	59.02
15	17	17	15	91.06	81.64	49.90
16	18	17	15	87.94	81.67	59.80
17	29	29	22	72.76	79.89	57.45
18	30	29	26	81.6	78.97	66.61
19	32	31	26	85.82	81.88	61.03
20	28	28	22	73.18	65.21	70.76
21	27	26	23	81.23	70.66	61.92
22	29	29	22	72.58	90.75	69.62
23	25	26	20	68.61	80.67	56.67
24	25	24	21	80.94	67.13	60.70
avg	23.50	23.13	19.19	75.50	75.70	59.82

are eliminated severally to derive the dominance percentage of each method.

$$Per(S_i) = \frac{|S_i - \{x \in S_i | \exists z \in S : z \prec x\}|}{|S_i|} \quad (30)$$

TABLE VI gives the results of cardinality and dominance percentage of MO-ALNS, ALNS- ϵ and NSGA-II, with fifteen runs of each method. As shown in TABLE VI, ALNS- ϵ and MO-ALNS both can generate more Pareto-optimal solutions and achieve higher dominance percentages compared to NSGA-II. ALNS- ϵ is capable of producing slightly more Pareto-optimal solutions than MO-ALNS on average. MO-ALNS achieved a mildly better performance according to the average domination percentage shown in the last row.

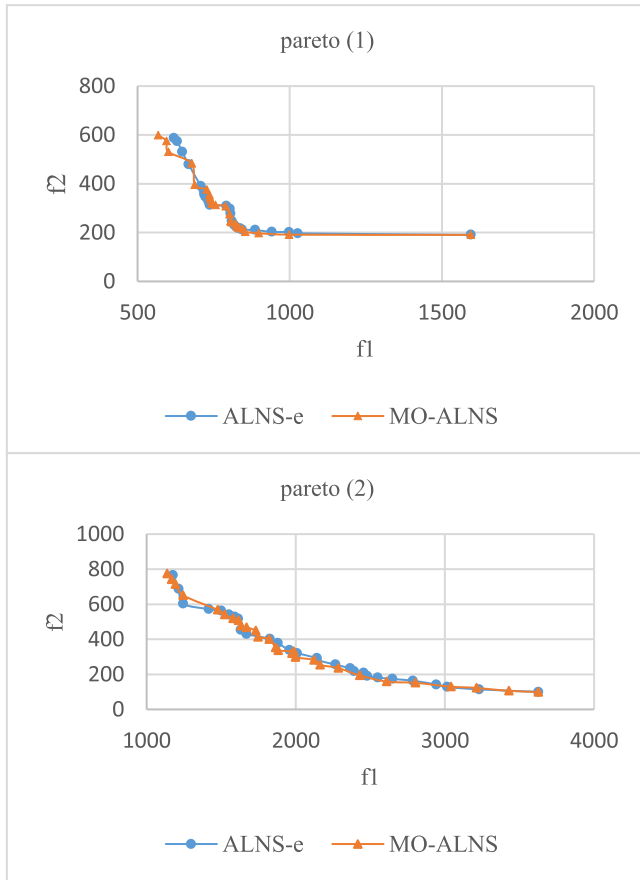
The graphic representation of the Pareto fronts obtained through ALNS- ϵ and MO-ALNS are depicted in Fig.3. Fig.3

shows that transportation cost and customer satisfaction are a pair of competing objectives that need to be traded off. For the same demand from customers, decision makers can adopt multiple delivery plans. Decision makers need to weigh balance between these two objectives. It can also be observed from Fig.3 that when the cost level is relatively low, the effect of increasing costs on improving customer satisfaction is more significant. In situations where the cost level is high, customer satisfaction has gradually approached saturation and the upward potential is limited.

Furthermore, to evaluate how the flexible departure time will affect the final solutions, TABLE VII compares the two situations where all vehicles depart at time zero and the departure time of vehicles is flexible. Two indexes are analyzed including the percentage of travel time reduction and the percentage of quality loss reduction. The reduction of travel time is calculated according to the average time duration of all routes. Instance 9,10,13,14,17,18,21,22 are geographically centralized distributed instances (Solomon-C type) and the others are geographically random-centralized distributed instances (Solomon-RC type).

TABLE VII shows that allowing flexible departure time considerably improves the solution quality, with the travel time reduction of 27.63% and product quality loss reduction of 33.97% on average. We observe that delayed departure of a vehicle can effectively reduce its time spent between some pair of customers without violating the delivery time window specified by customers, and as a result, the quality loss is reduced.

In addition, we notice that, when customers are more densely scattered geographically, delaying vehicle departure time performs more effectively in reducing delivery time and improving customer satisfaction. When all vehicles have a departure time of zero, a centralized customer distribution results in shorter travel times between two consecutive customers but potentially longer waiting times. Conversely, a dispersed customer distribution leads to longer travel times

Fig. 3. Pareto fronts obtained by MO-ALNS and ALNS- ϵ .TABLE VII
EFFECTS OF FLEXIBLE DEPARTURE TIME

Instance	Time reduction	Quality loss reduction
9	0.317868	0.47069
10	0.261226	0.347945
11	0.188833	0.21994
12	0.148878	0.216469
13	0.433815	0.600714
14	0.349845	0.465879
15	0.328991	0.357345
16	0.291648	0.380777
17	0.368014	0.369741
18	0.286017	0.336350
19	0.155716	0.137401
20	0.155014	0.125709
21	0.516799	0.665233
22	0.409845	0.440436
23	0.123758	0.189520
24	0.085226	0.111379
avg	0.276343	0.339721

between consecutive customers but reduced waiting times. Therefore, delaying the departure of vehicles has a more significant impact on centralized instances.

Moreover, the customers in instance 21 and 22 all choose the first delivery mode whereas the customers in instance 23 and 24 all choose the second delivery mode. The results show that as the proportion of the first type of customers increases, the effect of the flexible departure time becomes more significant, this may be because shorter delivery time window will

TABLE VIII
EFFECT OF POST-PROCESSING

Instance	MO-ALNS		ALNS- ϵ	
	Number of solutions	Improved solutions	Number of solutions	Improved solutions
9	18	6	18	6
10	21	5	20	5
11	20	6	20	7
12	20	4	19	3
13	19	6	19	6
14	18	6	18	6
15	17	4	17	4
16	18	4	17	5
17	29	8	29	9
18	30	9	29	8
19	32	6	31	6
20	28	5	28	5
21	27	4	26	3
22	29	8	29	9
23	25	7	26	9
24	25	7	24	6
average	23.50	5.94	23.13	6.06

cause increased waiting time generally. We can also obtain a managerial insight that suppliers should strategically delay the departure of vehicles for delivering perishable products to customers. This delay becomes particularly advantageous when customers are located nearby.

3) Effectiveness of the Post-Processing Procedure:

Table VIII presents the number of solutions that have been improved among all the solutions obtained for each instance by the post-processing procedure, calculated by the sum of the dominated solution elimination. It suggests that dominated solutions can be avoided effectively by applying this method, thus it helps save the computation time.

C. Sensitivity Analysis

This subsection involves a sensitivity analysis aimed at investigating the effects of different parameters of the problem on its solutions.

1) *Different Proportions of Customers:* Firstly, we varied the proportion of the two types of customers in our problem to study its impact on the objective values. This proportion is represented by two numbers separated by “-” with their sum equal to 100. For example, 25-75 denotes that 25% of the customers choose the first delivery mode and the other 75% choose the second delivery mode. To examine the impact, two groups of instances are taken into consideration with $N = 100$ for each and the instances in each group are generated with five different proportion settings, which respectively are 0-100, 25-75, 50-50, 75-25, 100-0. Fig.4 illustrates the pareto fronts obtained for these two groups of instances.

As can be observed from Fig.4, it is evident that as the proportion of the first customers increases while preserving the customer satisfaction at the same level, the transportation cost increases. The higher percentage of the first type customers implies higher percentage of tighter time windows, reducing supplier flexibility in route planning. As a result, more vehicles are needed to meet customer demand, ultimately leading to increased costs.

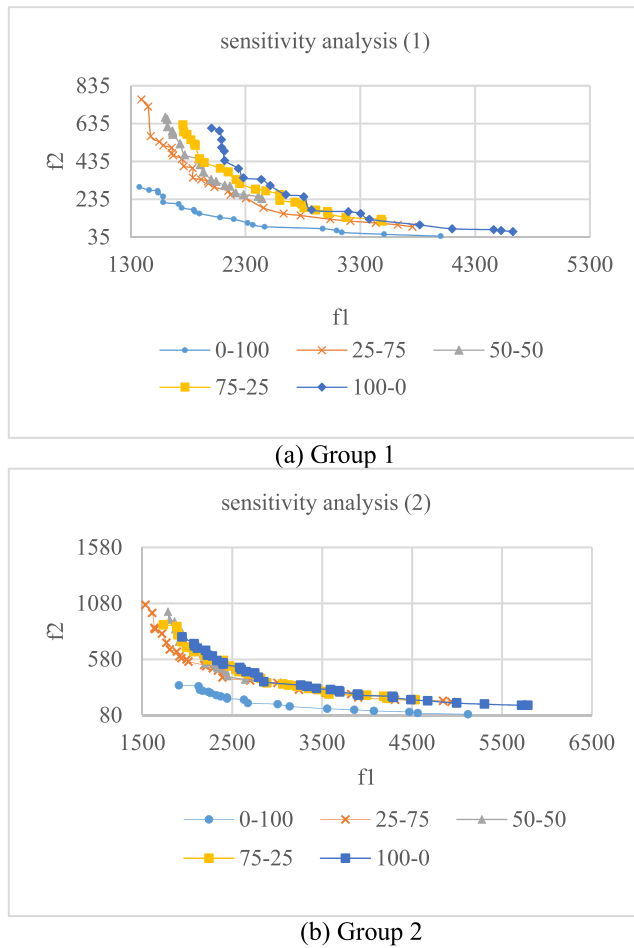


Fig. 4. Solutions with different proportions of customers.

2) *Different Tightness Levels of Delivery Time Windows:* Secondly, we consider how different tightness levels of delivery time windows will impact the solution of the studied problem. Two groups of instances are generated. For each group, we consider two settings of time window lengths: (i) 30min; and (ii) 60min. The results are given in Fig. 5 as follows.

As can be observed from Fig. 5, the overall trend is that customer satisfaction level is lower when the time window is strictly shorter under the same transportation cost. According to our analysis, shortening the time windows leads to a reduction in the number of customers that a single route can serve. Consequently, more vehicles are required to complete all delivery tasks and maintain the same level of customer satisfaction, resulting in increased transportation costs. Therefore, suppliers need to strike a balance between time window length and cost while providing customers with different options.

D. Case Study

In real distribution of perishable products, customer orders may dynamically arrive, leading to a variant of dynamic vehicle routing problem. The Dynamic Vehicle Routing Problem (DVRP) was first introduced by Wilson and Colvin [63]. DVRP enables the adjustment of vehicle routes based on

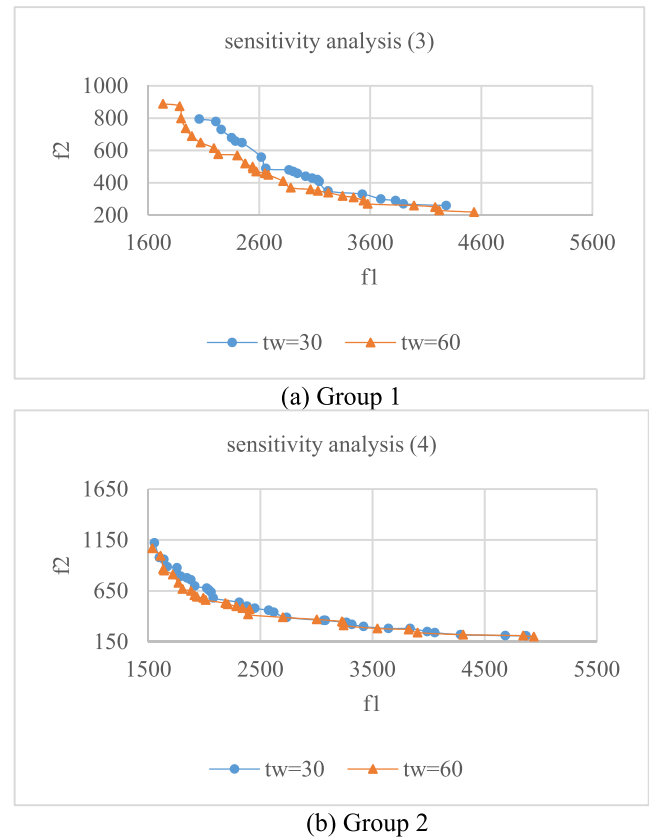


Fig. 5. Solutions with different tightness levels of time windows.

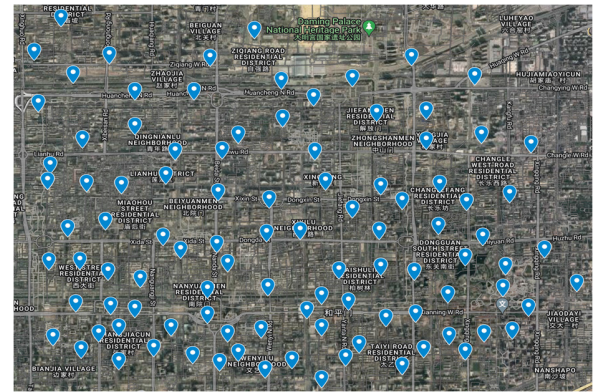


Fig. 6. Network of the case.

dynamically revealed information during operations. Our proposed ALNS method can address the dynamic arrival of customer orders by treating newly arrived orders as previously removed ones which can then be inserted into undispached vehicles to form updated delivery routes. To demonstrate this, a case study is conducted for a supplier operating in a logistics network located in Xi'an, China. Fig. 6 shows the network, which consists of 100 customer nodes and a depot of the supplier.

The supplier has 25 vehicles with a capacity of 200 and the planning horizon is set to 4 hours. For the scenario considered, forty customers' demands are known in advance, with additional twenty new customers' demands to arrive every

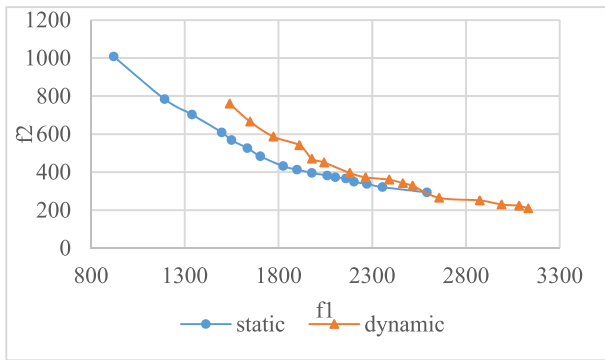


Fig. 7. Results of the case study.

hour. Fig. 7 illustrates the computational results of the scenario with dynamic arrival of customer demands and the static scenario with all customer demands known in advance. This figure indicates that our proposed algorithm can deal with the dynamic scenario. The maximum customer satisfaction level achieved is higher in the dynamic scenario compared to the static scenario. However, achieving the same level of customer satisfaction in the dynamic scenario incurs higher costs. This may be because, compared to the static scenario, there is a higher probability that newly arrived orders are assigned to new vehicles in the dynamic scenario, which may lead to more vehicles used to deliver all orders with lower routing efficiency and higher delivery costs.

VI. CONCLUSION

The motivation behind this study stems from the facts that the distribution of perishable products has become a large part of road freight transportation with increasing demands, the consideration of both transportation costs and customer satisfaction is necessary for effective distribution of such products in order to increase market share, and customers may have different preferences for delivery time windows. Apart from these, no study has considered customers' priorities in the context of differentiated time windows in the literatures regarding BO-VRP for perishable products as well as the flexible vehicle departure time. The goal of our study is to bridge this research gap.

A bi-objective VRP optimization model for perishable products was formulated. To cope with the complexity of the considered problem, we developed two meta-heuristic methods which are ALNS- ϵ and MO-ALNS. Computational experiments were conducted to compare ALNS- ϵ , MO-ALNS, and NSGA-II. The results of the experiments demonstrate the effectiveness of the two algorithms proposed in this paper. Moreover, a sensitivity analysis and case study are performed. Besides, allowing flexible vehicle departure time can reduce delivery time and promote customer satisfaction. With regard to the limitations, while our model is designed to be broadly applicable to perishable goods and it can be adapted to a wide range of perishable goods exhibiting linear deterioration. Further consideration can be given to other types of perishable goods to enhance the model generalizability. In addition, the algorithm may require finer parameter tuning or the

implementation of specific optimization strategies for other specific types of perishable goods. The integration of diverse metaheuristics into a multi-objective optimization framework is also an extended direction to overcome the limitation of only considering the integration of the ALNS. Furthermore, there are limitations in considering dynamic factors, and more dynamic factors could be considered.

In future, the current work can be extended by integrating additional metaheuristic components into the multi-objective optimization framework to further enhance its performance. Apart from this, dynamic and stochastic elements can be further considered to dynamically optimize the delivery plan for a perishable product supplier, taking into account real-time traffic conditions and dynamic order arrivals. Furthermore, our work can be extended to apply to a broader range of perishable goods with varying characteristics.

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